

A Cohort formation

OAI development cohort

9,592 knees / 4,796 participants
Longitudinal symptoms and centrally read radiographs

Sequential exclusions

Baseline/prior KR: 69; target-knee KR before or at landmark: 74
No post-landmark follow-up: 692; missing landmark pain: 443
Missing one or more Model F-core predictors: 5,210

Model F-core

3,104 knees / 1,656 participants
566 post-landmark KR events

Model E

3,066 knees / 1,640 participants
569 post-landmark KR events

MRKR transport cohort

9,115 knee-level pairs / 5,162 patients
Real-world radiographs and linked clinical records

Sequential exclusions

No valid post-landmark hardware-observation interval: 4,506
Missing landmark pain: 1,197

Model F-core transport cohort

3,412 knees / 2,179 patients
1,140 side-specific hardware detections; 855 by 24 months

B Twenty-four-month landmark design



C Distinct modeling roles

